

KMTNet Auxiliary control software – Remote commands definition

1. Protocol and configuration

- 1) Communication type: TCP/IP (Server: Auxiliary GUI control SW, Client: Remote user)
 - 2) Message packet type: Message string (ASCII character array)
 - 3) Server IP address: 192.168.24.10
 - 4) Port number: 5752
- * The IP address and the port number can be changed on the server setting GUI of AUX control software and on the network adapter setting of AUX control computer.

2. Command message format

- 1) Message string format
 - Consist of 4 or more arguments, parsed by splitting at the spaces (0x20 in ASCII)
 - [LF] terminator is appended at the end of string. ([LF] = ' \ n' = 0x0A in ASCII)
- 2) Command message format
`<Telescope ID> <System> <Packet ID> <SUBSYSTEM> <COMMAND>[LF]`
- 3) Response message format
`<Telescope ID> <System> <Packet ID> <RESPONSE>[LF]`
- 4) Arguments description
 - `<Telescope ID>` : customized in Server Setting dialog (default: "KMTNET")
 - `<System>` : hardcoded to "AUX", for auxiliary GUI control SW
 - `<Packet ID>` : any numeric or string value that does not contain a space
 - `<SUBSYSTEM>` : the subsystem that applies to the command
 - `<COMMAND>` : the command to send to the subsystem

* If `<Telescope ID>` or `<System>` is incorrect, or the arguments number is insufficient, the server does NOT return any response.
- 5) Example


```
KMTNET AUX 123 FOCUSER CONNECT
KMTNET AUX 123 SUCCESS

KMTNET TCS 123 FUCUSER CONNECT → (No response)
KMTN AUX 123 FUCUSER CONNECT → (No response, if Telescope ID is "KMTNET")
```

3. Command description

1) Command list

Subsystem	Command	Parameters	Response	Notes	Idx
ALL	CONNECT		<Result>	All subsystems connection	1-1
	DISCONNECT		<ACK>	All subsystems disconnection	1-2
	STATUS		<All statuses>	Statuses for all subsystems	1-3
	ECHO	<Echo message>	<Echo message>	For communication check	1-4
common	CONNECT		<Result>	Connection to the device	1-5
	DISCONNECT		<ACK>	Disconnection from the device	1-6
	STATUS		<Status>	Status of each subsystem	1-7
	responses		<ACK> / <NACK> / <WAIT> / <ERROR>	→ refer to table 1-0	1-0
FOCUSER	LIMIT		<AL1> <AL2> <AL3>	Status of both out & in limits	2-1
	POSITION		<AP1> <AP2> <AP3>	"±0.00 ±0.00 ±0.00", mm	2-2
	GOTO_A1	<A1>	<ACK> / <WAIT>	"±00.000", mm	2-3
	GOTO_A2	<A2>	<ACK> / <WAIT>	"±00.000", mm	2-3
	GOTO_A3	<A3>	<ACK> / <WAIT>	"±00.000", mm	2-3
	GOTO_ALL	<A1> <A2> <A3>	<ACK> / <WAIT>	"±00.000 ±00.000 ±00.000"	2-3
	OFFSET	<Offset>	<ACK> / <WAIT>	"±00.000", mm	2-4
	HOME		<ACK> / <WAIT>	Home search and initialization	2-5
SHUTTER	ELEVATION		<Shut. Elev.> <Tele. Elev.>	"00.0", degree	3-1
	LIMITS		<Up> <Low> <Safe>	→ refer to table 3-2 for details	3-2
	BOTH	<Action>	<ACK> / <WAIT>	"OPEN" / "CLOSE"	3-3
	UPPER	<Action>	<ACK> / <WAIT>	"OPEN" / "CLOSE"	3-3
	LOWER	<Action>	<ACK> / <WAIT>	"OPEN" / "CLOSE"	3-3
	GOTO	<Elevation>	<ACK> / <WAIT>	"00.0" degree	3-4
	STOP		<ACK>	Stop operation	3-5
	SET_ASYNC	<Action>	<ACK> / <WAIT>	"ON" / "OFF"	3-6
	GET_ASYNC		<AutoSync>	"ON" / "OFF"	3-6
FILTERS	LIMIT_FILTER		<FL1> <FL2> <FL3> <FL4>	Status of all filter limit switches	4-1
	LIMIT_SHUT		<SL1> <SL2>	Status of all shutter limit	4-2
	SET_F1	<Action>	<ACK> / <WAIT>	"IN" / "OUT"	4-3
	SET_F2	<Action>	<ACK> / <WAIT>	"IN" / "OUT"	4-3
	SET_F3	<Action>	<ACK> / <WAIT>	"IN" / "OUT"	4-3
	SET_F4	<Action>	<ACK> / <WAIT>	"IN" / "OUT"	4-3
	FNAMES		<FN1> <FN2> <FN3> <FN4>	Filter names	4-4
M1COVER	POSITION		<Mirror cover position>	"0" ~ "100", Percentage	5-1
	OPEN		<ACK> / <WAIT>	Start mirror cover opening	5-2
	CLOSE		<ACK> / <WAIT>	Start mirror cover closing	5-2
	STOP		<ACK>	Stop operation	5-3
CHILLER	GET_TEMP		<Proc. temperature>	"±0.0", °C	6-1
	SET_SETPOINT	<Set point>	<ACK>	"±0.0", °C	6-2

	GET_SETPOINT		<Set point>	"±0.0", °C	6-2
	SET_POWER	<Action>	<ACK>	"ON" / "OFF"	6-3
	GET_POWER		<Chiller power>	"ON" / "OFF"	6-3
ENVIRON	SENSORS		<SE1> <SE2> ... <SE7>	"0.0 0.0 ... 0.0", °C/RH%	6-4
	SET_FAN	<Action>	<ACK>	"ON" / "OFF"	6-5
	GET_FAN		<Fan power>	"ON" / "OFF"	6-5

2) Command details

A. Subsystem ALL: Master control and Common command/response

1-0. common – response			
Response	String	Descriptions	Related CMD
<ACK>	"OK"	Acknowledgement message according to receiving proper commands	All commands that don't request parameter return
<NACK>	"BAD"	Negative-Acknowledgment message * In the case below, <NACK> is returned. - Unrecognized commands or arguments number - Invalid parameter number or range - Unavailable operating command due to HW status - commands for operating the lower dome shutter when the safety interlock is active - GOTO command in the Auto-sync mode - Commands except CONNECT, DISCONNECT and STATUS when the subsystem is not connected	All invalid commands and arguments, and (3-3) LOWER (3-4) GOTO
<WAIT>	"WAIT"	Notification that the subsystem is currently unable to start the operation since previous operation is not completed yet * In the case below, <WAIT> is returned. - Commands for operating Focuser when any focuser actuator is running - BOTH/UPPER/LOWER/GOTO/SET_ASYNC commands for operating the dome shutter when the Dome Shutter is Opening/Closing - SET_Fx command when commanded filter slide	7 Commands for mechanical operation that need some operation time for the completion: (2-3) GOTO_Ax (2-4) OFFSET (2-5) HOME (3-3) BOTH

		(filter number x) is running - OPEN/CLOSE Commands for Opening/Closing the mirror cover when they are already running * Although previous GOTO operation is in progress, newly commanded BOTH/UPPER/LOWER/GOTO/SET_ASYNC operation will be started after stopping previous GOTO operation automatically.	(3-3) UPPER (3-3) LOWER (3-4) GOTO (3-6) SET_ASYNC (4-5) SET_Fx (5-2) OPEN (5-2) CLOSE
<ERROR>	"ERROR"	Notification that commanded operation is failed or an error occurred in previous operation * In case ERROR, <Status> is also set to ERROR.	All commands except CONNECT
<Result>	"SUCCESS" "FAILURE"	The operation result that is returned when after the operation is completed	CONNECT
<Status>	"NC" "STANDBY" "RUNNING" "ERROR"	Operational status of each subsystem - NC: The subsystem is not connected to the device. - STANDBY: The subsystem is connected, standby, and waiting a command. - RUNNING: The subsystem is running for mechanical operation such as Focuser actuator, Filter slide, Dome Shutter and Mirror cover. - ERROR: Operation error occurred during previous operation of the subsystem. ERROR status changes to STANDBY status when the subsystem successfully completes another operation.	STATUS

1-1. ALL – CONNECT	
<SUBSYSTEM>	"ALL"
<COMMAND>	"CONNECT"
<RESPONSE>	<Result>
Parameter	-
Function	All subsystems connect to each device.
Example	KMTNET AUX A01 ALL CONNECT KMTNET AUX A01 SUCCESS KMTNET AUX B01 ALL CONNECT KMTNET AUX B01 FAILURE KMTNET AUX B02 ALL STATUS KMTNET AUX B02 FOCUSER ERROR 0 0 0 +0.000 +0.000 +0.000 SHUTTER STANDBY 17.0 89.9 CLOSED CLOSED ACTIVE OFF FILTERS STANDBY 1 1 1 1 2 1 M1COVER STANDBY 0 CHILLER STANDBY +0.0 +5.0 OFF ENVIRON

	STANDBY 15.9 16.4 16.3 19.9 15.5 15.7 0.0 OFF
Notes	-

1-2. ALL – DISCONNECT	
<SUBSYSTEM>	"ALL"
<COMMAND>	"DISCONNECT"
<RESPONSE>	<ACK>
Parameter	-
Function	All subsystems disconnect from each device.
Example	<pre> KMTNET AUX 123 ALL STATUS KMTNET AUX 123 FOCUSER STANDBY 0 0 0 +0.111 +0.222 +0.333 SHUTTER STANDBY 17.0 89.9 CLOSED CLOSED ACTIVE OFF FILTERS STANDBY 1 1 1 1 2 1 M1COVER STANDBY 0 CHILLER STANDBY +0.0 +5.0 OFF ENVIRON STANDBY 15.9 16.4 16.3 19.9 15.5 15.7 0.0 OFF KMTNET AUX 456 ALL DISCONNECT KMTNET AUX 456 OK KMTNET AUX 789 ALL STATUS KMTNET AUX 789 FOCUSER NC 0 0 0 +0.111 +0.222 +0.333 SHUTTER NC 17.0 89.9 CLOSED CLOSED ACTIVE OFF FILTERS NC 1 1 1 1 2 1 M1COVER NC 0 CHILLER NC +0.0 +5.0 OFF ENVIRON NC 15.9 16.4 16.3 19.9 15.5 15.7 0.0 OFF </pre>
Notes	-

1-3. ALL – STATUS	
<SUBSYSTEM>	"ALL"
<COMMAND>	"STATUS"
<RESPONSE>	<All statuses>
Parameter	- <All statuses> : All statuses and values of all the subsystems, The string of <All statuses> response consists of the following arguments. <pre> 'FOCUSER' <Status> <AL1> <AL2> <AL3> <AP1> <AP2> <AP3> 'SHUTTER' <Status> <Shut. Elev.> <Tele. Elev.> <Up> <Low> <Safe> <AutoSync> 'FILTERS' <Status> <FL1> <FL2> <FL3> <FL4> <SL1> <SL2> 'M1COVER' <Status> <Mirror cover position> 'CHILLER' <Status> <Proc. temperature> <Set point> <Chiller power> 'ENVIRON' <Status> <SE1> <SE2> <SE3> <SE4> <SE5> <SE6> <SE7> <Fan power> </pre>
Function	Monitoring operational statuses and values for all the subsystems
Example	KMTNET AUX 123 ALL STATUSES

	KMTNET AUX 123 FOCUSER STANDBY 0 0 0 +1.111 +1.222 +1.333 SHUTTER STANDBY 60.0 60.0 MID OPEN INACTIVE ON FILTERS STANDBY 2 1 1 1 2 1 M1COVER STANDBY 100 CHILLER STANDBY +10.2 +5.0 OFF ENVIRON STANDBY 13.2 13.6 13.7 20.4 15.5 15.6 0.0 OFF
Notes	- The name of subsystems is at the head of status and values for the each subsystem in the middle of <All statuses> response.

1-4. ALL – ECHO	
<SUBSYSTEM>	"ALL"
<COMMAND>	"ECHO" <Echo message>
<RESPONSE>	<Echo message>
Parameter	- <Echo message>: User-defined message for communication check, The echo message string should not contain a space (0x20 in ASCII).
Function	Communication check with echo message
Example	KMTNET AUX 123 ALL ECHO check_message KMTNET AUX 123 check_message
Notes	-

1-5. common – CONNECT	
<SUBSYSTEM>	Common to all subsystems
<COMMAND>	"CONNECT"
<RESPONSE>	<Result>
Parameter	-
Function	The subsystem connects to the device.
Example	KMTNET AUX F001 FOCUSER CONNECT KMTNET AUX F001 SUCCESS KMTNET AUX S001 SHUTTER CONNECT KMTNET AUX S001 SUCCESS
Notes	-

1-6. common – DISCONNECT	
<SUBSYSTEM>	Common to all subsystems
<COMMAND>	"DISCONNECT"
<RESPONSE>	<ACK>

Parameter	-
Function	The subsystem disconnects from the device.
Example	KMTNET AUX F002 FOCUSER DISCONNECT KMTNET AUX F002 OK KMTNET AUX S002 SHUTTER DISCONNECT KMTNET AUX S002 OK
Notes	-

1-7. common – STATUS	
<SUBSYSTEM>	Common to all subsystems
<COMMAND>	"STATUS"
<RESPONSE>	<Status>
Parameter	- <Status> : There are 4 types of operational status as below - "NC" : The subsystem is not connected to the device. - "STANDBY" : The subsystem is connected and waiting remote command. - "RUNNING" : The subsystem is running for mechanical operations such as Focuser actuator, Filter slide, Dome Shutter and Mirror cover. - "ERROR" : Operational error occurred during previous operation of the subsystem. ERROR status changes to STANDBY status when the subsystem successfully completes another operation.
Function	Monitoring the operational status of the subsystem
Example	KMTNET AUX 001 FOCUSER STATUS KMTNET AUX 001 NC KMTNET AUX 002 FOCUSER CONNECT KMTNET AUX 002 SUCCESS KMTNET AUX 003 FOCUSER STATUS KMTNET AUX 003 STANDBY KMTNET AUX 004 FOCUSER HOME KMTNET AUX 004 OK KMTNET AUX 005 FOCUSER GOTO_ALL 1.1 2.2 3.3 KMTNET AUX 005 WAIT KMTNET AUX 006 FOCUSER STATUS KMTNET AUX 006 RUNNING KMTNET AUX 007 FOCUSER STATUS KMTNET AUX 007 RUNNING KMTNET AUX 008 FOCUSER STATUS KMTNET AUX 008 STANDBY KMTNET AUX 009 FOCUSER GOTO_ALL 3.1 3.2 3.3 KMTNET AUX 009 OK KMTNET AUX 010 FOCUSER DISCONNECT KMTNET AUX 010 OK KMTNET AUX 011 FOCUSER STATUS KMTNET AUX 011 NC
Notes	- This command could be useful for a remote-user to monitor either the

	operation is complete or still running, and to check connection or error status.
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B. Subsystem FOCUSER: Trio-actuators control for Head-ring tip/tilt/focusing

2-1. FOCUSER – LIMIT				
<SUBSYSTEM>	"FOCUSER"			
<COMMAND>	"LIMIT"			
<RESPONSE>	<AL1> <AL2> <AL3>			
Parameter	- <AL1> ~ <AL3>: Status of both Inner limit and Outer limit for 3 actuators	Outer limit	Inner limit	Status <AL>
		OFF	OFF	"0"
		ON	OFF	"1"
		OFF	ON	"2"
		ON	ON	"3"
Function	Monitoring HW limits of three actuators			
Example	KMTNET AUX 01 FOCUSER LIMIT KMTNET AUX 01 0 0 0 KMTNET AUX 02 FOCUSER GOTO_ALL +20.0 +20.0 +20.0 KMTNET AUX 02 OK KMTNET AUX 03 FOCUSER STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 FOCUSER STATUS KMTNET AUX 04 STANDBY KMTNET AUX 05 FOCUSER LIMIT KMTNET AUX 05 1 1 1			
Notes	-			

2-2. FOCUSER – POSITION	
<SUBSYSTEM>	"FOCUSER"
<COMMAND>	"POSITION"
<RESPONSE>	<AP1> <AP2> <AP3>
Parameter	- <AP1> ~ <AP3>: Calculated millimeter positions of 3 actuators, consist of 3 arguments, floating-point number including a sign. → ex) +12.345
Function	Monitoring the positions of three actuators
Example	<pre> KMTNET AUX 01 FOCUSER POSITION KMTNET AUX 01 +0.000 +0.000 +0.000 KMTNET AUX 02 FOCUSER OFFSET +10.0 </pre>

	KMTNET AUX 02 OK KMTNET AUX 03 FOCUSER POSITION KMTNET AUX 03 +4.384 +4.384 +4.384 KMTNET AUX 04 FOCUSER POSITION KMTNET AUX 04 +6.561 +6.561 +6.561 KMTNET AUX 05 FOCUSER POSITION KMTNET AUX 05 +10.000 +10.000 +10.000
Notes	-

2-3. FOCUSER – GOTO_A1 ~ GOTO_ALL	
<SUBSYSTEM>	"FOCUSER"
<COMMAND>	"GOTO_A1" <A1> "GOTO_A2" <A2> "GOTO_A3" <A3> "GOTO_ALL" <A1> <A2> <A3>
<RESPONSE>	<ACK > / <WAIT>
Parameter	- <A1>, <A2>, <A3>: Command position in millimeter for each actuator, floating-point number including a sign. → ex) "+12.345"
Function	Moving the actuators to the command positions
Example	KMTNET AUX 01 FOCUSER POSITION KMTNET AUX 01 -0.100 +0.200 -0.300 KMTNET AUX 02 FOCUSER GOTO_A1 -0.4 KMTNET AUX 02 OK KMTNET AUX 03 FOCUSER POSITION KMTNET AUX 03 -0.400 +0.200 -0.300 KMTNET AUX 04 FOCUSER GOTO_ALL 0.5 0.6 0.7 KMTNET AUX 04 OK KMTNET AUX 05 FOCUSER POSITION KMTNET AUX 05 +0.500 +0.600 +0.700 KMTNET AUX 11 FOCUSER GOTO_ALL 10.0 10.0 10.0 KMTNET AUX 11 OK KMTNET AUX 12 FOCUSER GOTO_ALL 0.0 0.0 0.0 KMTNET AUX 12 WAIT KMTNET AUX 13 FOCUSER STATUS KMTNET AUX 13 RUNNING KMTNET AUX 14 FOCUSER STATUS KMTNET AUX 14 STANDBY KMTNET AUX 15 FOCUSER POSITION KMTNET AUX 15 +10.000 +10.000 +10.000 KMTNET AUX 16 FOCUSER GOTO_ALL 0.0 0.0 0.0 KMTNET AUX 16 OK
Notes	- If any focuser actuator is still running for previous operation, commanded GOTO_Ax operation is not started and <WAIT> is returned. - The range of actuator position is around ± 10 mm. - The ranges of each actuator depend on the HW limit (factory setting).

2-4. FOCUSER – OFFSET	
<SUBSYSTEM>	"FOCUSER"
<COMMAND>	"OFFSET" <Offset>
<RESPONSE>	<ACK > / <WAIT>
Parameter	- <Offset>: Offset position in millimeter from the current positions of all actuators, floating-point number including a sign. → ex) "+12.345"
Function	Moving all actuators as commanded offset from the current position
Example	<pre> KMTNET AUX 01 FOCUSER POSITION KMTNET AUX 01 -0.100 +0.200 -0.300 KMTNET AUX 02 FOCUSER OFFSET 1.0 KMTNET AUX 02 OK KMTNET AUX 03 FOCUSER POSITION KMTNET AUX 03 +0.900 +1.200 +0.700 KMTNET AUX 04 FOCUSER OFFSET 1.0 KMTNET AUX 04 OK KMTNET AUX 05 FOCUSER POSITION KMTNET AUX 05 +1.900 +2.200 +1.700 KMTNET AUX 01 FOCUSER OFFSET -10.0 KMTNET AUX 01 OK KMTNET AUX 02 FOCUSER OFFSET +2.0 KMTNET AUX 02 WAIT KMTNET AUX 03 FOCUSER STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 FOCUSER STATUS KMTNET AUX 04 STANDBY KMTNET AUX 05 FOCUSER OFFSET +2.0 KMTNET AUX 05 OK </pre>
Notes	- If any focuser actuator is still running for previous operation, commanded OFFSET operation is not started and <WAIT> is returned.

2-5. FOCUSER – HOME	
<SUBSYSTEM>	"FOCUSER"
<COMMAND>	"HOME"
<RESPONSE>	<ACK > / <WAIT>
Parameter	-
Function	Home search and initialization of all actuators
Example	<pre> KMTNET AUX 0 FOCUSER POSITION KMTNET AUX 0 +1.900 +2.200 +1.700 KMTNET AUX 1 FOCUSER HOME KMTNET AUX 1 OK KMTNET AUX 2 FOCUSER GOTO_ALL 1.1 2.2 3.3 </pre>

	KMTNET AUX 2 WAIT KMTNET AUX 3 FOCUSER STATUS KMTNET AUX 3 RUNNING KMTNET AUX 4 FOCUSER STATUS KMTNET AUX 4 RUNNING KMTNET AUX 5 FOCUSER STATUS KMTNET AUX 5 STANDBY KMTNET AUX 6 FOCUSER POSITION KMTNET AUX 6 +0.000 +0.000 +0.000 KMTNET AUX 7 FOCUSER GOTO_ALL 1.1 2.2 3.3 KMTNET AUX 7 OK KMTNET AUX 8 FOCUSER POSITION KMTNET AUX 8 +1.100 +1.200 +1.300
Notes	- If any focuser actuator is still running for previous operation, commanded HOME operation is not started and <WAIT> is returned. - The initialization positions can be defined on the GUI SW setting dialog by on-site user.

C. Subsystem SHUTTER: Dome Shutter control

3-1. SHUTTER – ELEVATION	
<SUBSYSTEM>	"SHUTTER"
<COMMAND>	"ELEVATION"
<RESPONSE>	<Shut. Elev.> <Tele. Elev.>
Parameter	- <Shut. Elev.>: Current elevation of the upper dome shutter, floating-point number in degree → e.g. "65.5" - <Tele. Elev.>: Current elevation of the telescope, received from TCS, floating-point number in degree → e.g. "65.5"
Function	Monitoring the elevation angles of the upper dome shutter and the telescope
Example	KMTNET AUX 01 SHUTTER ELEVATION KMTNET AUX 01 60.0 70.1 KMTNET AUX 02 SHUTTER GOTO 70.0 KMTNET AUX 02 OK KMTNET AUX 03 SHUTTER STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 SHUTTER ELEVATION KMTNET AUX 04 63.4 70.1 KMTNET AUX 05 SHUTTER ELEVATION KMTNET AUX 05 66.8 70.1 KMTNET AUX 06 SHUTTER ELEVATION KMTNET AUX 06 70.0 70.1 KMTNET AUX 07 SHUTTER STATUS KMTNET AUX 07 STANDBY

Notes	-
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3-2. SHUTTER – LIMITS	
<SUBSYSTEM>	"SHUTTER"
<COMMAND>	"LIMITS"
<RESPONSE>	<Up> <Low> <Safe>
Parameter	- <Up> : Status of upper shutter limit switch → "OPEN" / "CLOSED" / "MID" - <Low>: Status of lower shutter limit switch → "OPEN" / "CLOSED" / "MID" - <Safe>: Status of safety interlock → "ACTIVE" / "INACTIVE"
Function	Monitoring the limit status of the dome shutters
Example	<pre> KMTNET AUX 01 SHUTTER LIMITS KMTNET AUX 01 CLOSED CLOSED ACTIVE KMTNET AUX 02 SHUTTER BOTH OPEN KMTNET AUX 02 OK KMTNET AUX 03 SHUTTER LIMITS KMTNET AUX 03 MID CLOSED ACTIVE KMTNET AUX 04 SHUTTER LIMITS KMTNET AUX 04 MID MID INACTIVE KMTNET AUX 05 SHUTTER STATUS KMTNET AUX 05 RUNNING KMTNET AUX 06 SHUTTER STATUS KMTNET AUX 06 STANDBY KMTNET AUX 07 SHUTTER LIMITS KMTNET AUX 07 OPEN OPEN INACTIVE </pre>
Notes	-

3-3. SHUTTER – BOTH, UPPER, LOWER	
<SUBSYSTEM>	"SHUTTER"
<COMMAND>	"BOTH" <Action> "UPPER" <Action> "LOWER" <Action>
<RESPONSE>	<ACK> / <WAIT>
Parameter	- <Action> → "OPEN" / "CLOSE"
Function	Starting the dome shutters Open/Close operation
Example	<pre> KMTNET AUX 01 SHUTTER LIMITS KMTNET AUX 01 OPEN OPEN INACTIVE KMTNET AUX 02 SHUTTER BOTH CLOSE KMTNET AUX 02 OK KMTNET AUX 03 SHUTTER STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 SHUTTER LIMITS </pre>

	KMTNET AUX 04 MID MID INACTIVE KMTNET AUX 05 SHUTTER LIMITS KMTNET AUX 05 MID CLOSED INACTIVE KMTNET AUX 06 SHUTTER LIMITS KMTNET AUX 06 MID CLOSED ACTIVE KMTNET AUX 07 SHUTTER STATUS KMTNET AUX 07 RUNNING KMTNET AUX 08 SHUTTER STATUS KMTNET AUX 08 RUNNING KMTNET AUX 09 SHUTTER STATUS KMTNET AUX 09 STANDBY KMTNET AUX 10 SHUTTER LIMITS KMTNET AUX 10 CLOSED CLOSED ACTIVE KMTNET AUX 11 SHUTTER LIMITS KMTNET AUX 11 CLOSED CLOSED ACTIVE KMTNET AUX 12 SHUTTER LOWER OPEN KMTNET AUX 12 BAD KMTNET AUX 13 SHUTTER UPPER OPEN KMTNET AUX 13 OK KMTNET AUX 14 SHUTTER LIMITS KMTNET AUX 14 MID CLOSED ACTIVE KMTNET AUX 15 SHUTTER LIMITS KMTNET AUX 15 MID CLOSED INACTIVE KMTNET AUX 16 SHUTTER LOWER OPEN KMTNET AUX 16 WAIT KMTNET AUX 17 SHUTTER STOP KMTNET AUX 17 OK KMTNET AUX 18 SHUTTER LOWER OPEN KMTNET AUX 18 OK KMTNET AUX 19 SHUTTER LIMITS KMTNET AUX 19 MID MID INACTIVE KMTNET AUX 18 SHUTTER UPPER OPEN KMTNET AUX 18 WAIT KMTNET AUX 19 SHUTTER STAUTS KMTNET AUX 19 RUNNING KMTNET AUX 20 SHUTTER STAUTS KMTNET AUX 20 STANDBY KMTNET AUX 21 SHUTTER LIMITS KMTNET AUX 21 MID OPEN INACTIVE KMTNET AUX 22 SHUTTER UPPER OPEN KMTNET AUX 22 OK KMTNET AUX 23 SHUTTER STAUTS KMTNET AUX 23 RUNNING KMTNET AUX 24 SHUTTER LIMITS KMTNET AUX 24 MID OPEN INACTIVE KMTNET AUX 25 SHUTTER LIMITS KMTNET AUX 25 OPEN OPEN INACTIVE KMTNET AUX 26 SHUTTER STAUTS KMTNET AUX 26 STANDBY
Notes	- If Dome Shutter is in the operation for Opening or Closing, OPEN/CLOSE operation cannot be started and <WAIT> is returned. - If Dome Shutter is in the GOTO operation, Even though it is not complete, newly commanded BOTH/UPPER/LOWER operation will be started after stopping previous GOTO operation automatically and <ACK> will be

	returned. - While Auto-Sync mode is enabled, if BOTH/UPPER/LOWER is commanded Shutter-positioning operation is stopped and Auto-Sync mode will be disabled and changed to Manual mode automatically. - If the upper shutter is almost closed and safety interlock is ACTIVE status, LOWER OPEN/CLOSE command is not available and the operation is not started and <BAD> is returned.
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3-4. SHUTTER – GOTO	
<SUBSYSTEM>	"SHUTTER"
<COMMAND>	"GOTO" <Elevation>
<RESPONSE>	<ACK> / <WAIT>
Parameter	- <Elevation>: Command elevation of the upper dome shutter in degree, floating-point number → ex) "65.5"
Function	Moving the upper dome shutter to the command elevation
Example	<pre> KMTNET AUX 01 SHUTTER ELEVATION KMTNET AUX 01 60.0 70.1 KMTNET AUX 02 SHUTTER GOTO 70.0 KMTNET AUX 02 OK KMTNET AUX 03 SHUTTER STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 SHUTTER STATUS KMTNET AUX 04 RUNNING KMTNET AUX 05 SHUTTER STATUS KMTNET AUX 05 STANDBY KMTNET AUX 06 SHUTTER ELEVATION KMTNET AUX 06 70.0 70.1 KMTNET AUX 07 SHUTTER ELEVATION KMTNET AUX 07 70.0 80.0 KMTNET AUX 08 SHUTTER GOTO 50.0 KMTNET AUX 08 OK KMTNET AUX 09 SHUTTER ELEVATION KMTNET AUX 09 62.0 80.0 KMTNET AUX 10 SHUTTER ELEVATION KMTNET AUX 10 60.0 80.0 KMTNET AUX 11 SHUTTER STATUS KMTNET AUX 11 RUNNING KMTNET AUX 12 SHUTTER GOTO 80.0 KMTNET AUX 12 OK KMTNET AUX 13 SHUTTER ELEVATION KMTNET AUX 13 62.0 80.0 KMTNET AUX 14 SHUTTER ELEVATION KMTNET AUX 14 64.0 80.0 KMTNET AUX 15 SHUTTER STATUS KMTNET AUX 15 RUNNING KMTNET AUX 16 SHUTTER STATUS </pre>

	KMTNET AUX 16 RUNNING KMTNET AUX 17 SHUTTER STATUS KMTNET AUX 17 STANDBY KMTNET AUX 18 SHUTTER ELEVATION KMTNET AUX 18 80.0 80.0 KMTNET AUX G01 SHUTTER LIMITS KMTNET AUX G01 CLOSED CLOSED ACTIVE KMTNET AUX G02 SHUTTER BOTH OPEN KMTNET AUX G02 OK KMTNET AUX G03 SHUTTER STATUS KMTNET AUX G03 RUNNING KMTNET AUX G04 SHUTTER GOTO 60.0 KMTNET AUX G04 WAIT KMTNET AUX G05 SHUTTER STOP KMTNET AUX G05 OK KMTNET AUX G07 SHUTTER STATUS KMTNET AUX G07 STANDBY KMTNET AUX G08 SHUTTER GOTO 60.0 KMTNET AUX G08 OK KMTNET AUX G09 SHUTTER STATUS KMTNET AUX G09 RUNNING KMTNET AUX G10 SHUTTER BOTH CLOSE KMTNET AUX G10 OK KMTNET AUX G13 SHUTTER LIMITS KMTNET AUX G13 MID MID INACTIVE KMTNET AUX G14 SHUTTER STATUS KMTNET AUX G14 RUNNING KMTNET AUX G15 SHUTTER STATUS KMTNET AUX G15 STANDBY KMTNET AUX G16 SHUTTER LIMITS KMTNET AUX G16 CLOSED CLOSED ACTIVE
Notes	- If Dome Shutter is in the operation for Opening or Closing, GOTO operation cannot be started and <WAIT> is returned. - If Dome Shutter is in the GOTO operation, Even though it is not complete, newly commanded GOTO operation will be started after stopping previous GOTO operation automatically and <ACK> will be returned. - While Auto-Sync mode is enabled, GOTO command is not available and if GOTO is commanded <BAD> will be returned. - Positioning error of Upper dome shutter is generally ± 0.1 degree.

3-5. SHUTTER – STOP	
<SUBSYSTEM>	"SHUTTER"
<COMMAND>	"STOP"
<RESPONSE>	<ACK>
Parameter	-
Function	Dome shutter operation stop

Example	KMTNET AUX 1 SHUTTER BOTH OPEN KMTNET AUX 1 OK KMTNET AUX 2 SHUTTER STATUS KMTNET AUX 2 RUNNING KMTNET AUX 3 SHUTTER BOTH CLOSE KMTNET AUX 3 WAIT KMTNET AUX 4 SHUTTER STOP KMTNET AUX 4 OK KMTNET AUX 5 SHUTTER STATUS KMTNET AUX 5 STANDBY KMTNET AUX 6 SHUTTER BOTH CLOSE KMTNET AUX 6 OK
Notes	-

3-6. SHUTTER – SET_ASYNC/GET_ASYNC	
<SUBSYSTEM>	"SHUTTER"
<COMMAND>	"SET_ASYNC" <Action> "GET_ASYNC"
<RESPONSE>	(SET) <ACK> / <WAIT> (GET) <AutoSync>
Parameter	- <Action> → "ON" / "OFF" - <AutoSync> → "ON" / "OFF"
Function	Auto-sync mode Enable/Disable control and monitoring
Example	KMTNET AUX 01 SHUTTER GET_ASYNC KMTNET AUX 01 OFF KMTNET AUX 02 SHUTTER SET_ASYNC ON KMTNET AUX 02 OK KMTNET AUX 03 SHUTTER GET_ASYNC KMTNET AUX 03 ON KMTNET AUX 04 SHUTTER GOTO 60.0 KMTNET AUX 04 BAD KMTNET AUX 05 SHUTTER AUTOSYNC OFF KMTNET AUX 05 OK KMTNET AUX 06 SHUTTER GET_ASYNC KMTNET AUX 06 OFF KMTNET AUX 07 SHUTTER GOTO 60.0 KMTNET AUX 07 OK KMTNET AUX 08 SHUTTER ELEVATION KMTNET AUX 08 60.0 70.0 KMTNET AUX 09 SHUTTER STATUS KMTNET AUX 09 STANDBY KMTNET AUX 10 SHUTTER SET_ASYNC ON KMTNET AUX 10 OK KMTNET AUX 11 SHUTTER STATUS KMTNET AUX 11 RUNNING KMTNET AUX 12 SHUTTER ELEVATION KMTNET AUX 12 64.3 70.0 KMTNET AUX 13 SHUTTER ELEVATION


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KMTNET AUX 13 67.5 70.0
KMTNET AUX 14 SHUTTER ELEVATION
KMTNET AUX 14 70.0 70.0
KMTNET AUX 15 SHUTTER STATUS
KMTNET AUX 15 STANDBY

KMTNET AUX 16 SHUTTER GET_ASYNC
KMTNET AUX 16 ON
KMTNET AUX 17 SHUTTER BOTH CLOSE
KMTNET AUX 17 OK
KMTNET AUX 18 SHUTTER GET_ASYNC
KMTNET AUX 18 OFF

KMTNET AUX 19 SHUTTER ELEVATION
KMTNET AUX 19 17.0 70.0
KMTNET AUX 20 SHUTTER LIMITS
KMTNET AUX 20 CLOSED CLOSED ACTIVE
KMTNET AUX 21 SHUTTER BOTH OPEN
KMTNET AUX 21 OK
KMTNET AUX 22 SHUTTER STATUS
KMTNET AUX 22 RUNNING
KMTNET AUX 23 SHUTTER SET_ASYNC ON
KMTNET AUX 23 WAIT
KMTNET AUX 24 SHUTTER STOP
KMTNET AUX 24 OK

KMTNET AUX 25 SHUTTER ELEVATION
KMTNET AUX 25 18.4 70.0
KMTNET AUX 26 SHUTTER LIMITS
KMTNET AUX 26 MID CLOSED ACTIVE
KMTNET AUX 27 SHUTTER STATUS
KMTNET AUX 27 STANDBY
KMTNET AUX 28 SHUTTER SET_ASYNC ON
KMTNET AUX 28 OK
KMTNET AUX 29 SHUTTER ELEVATION
KMTNET AUX 29 19.4 70.0
KMTNET AUX 30 SHUTTER LIMITS
KMTNET AUX 30 MID CLOSED ACTIVE
KMTNET AUX 31 SHUTTER ELEVATION
KMTNET AUX 31 20.8 70.0
KMTNET AUX 32 SHUTTER LIMITS
KMTNET AUX 32 MID MID INACTIVE
KMTNET AUX 33 SHUTTER ELEVATION
KMTNET AUX 33 20.8 70.0
KMTNET AUX 34 SHUTTER LIMITS
KMTNET AUX 34 MID OPEN INACTIVE

KMTNET AUX 35 SHUTTER ELEVATION
KMTNET AUX 35 22.4 70.0
KMTNET AUX 36 SHUTTER ELEVATION
KMTNET AUX 36 24.6 70.0
KMTNET AUX 37 SHUTTER STATUS
KMTNET AUX 37 RUNNING
KMTNET AUX 38 SHUTTER STATUS
KMTNET AUX 38 STANDBY
KMTNET AUX 39 SHUTTER ELEVATION
KMTNET AUX 39 70.1 70.0
KMTNET AUX 40 SHUTTER GET_ASYNC

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	KMTNET AUX 40 ON
Notes	- While Auto-sync mode is enabled, the upper dome shutter will move according to the telescope elevation automatically. - If Dome Shutter is in the GOTO operation, Even though it is not complete SET_ASYNC ON command is available and the operation will be started after stopping previous GOTO operation automatically and <ACK> will be returned. - If Dome Shutter is in the operation for Opening or Closing, SET_ASYNC command is unavailable and <WAIT> is returned. - While Auto-Sync mode is enabled, if BOTH/UPPER/LOWER is commanded Shutter-positioning operation is stopped and Auto-Sync mode will be disabled and changed to Manual mode automatically. - While Auto-sync mode is enabled, GOTO command is unavailable and the operation is not started and <BAD> is returned. - If Lower Shutter is closed when Auto-sync mode is enabled, the Auto-sync process will open Lower Shutter completely before start the process to monitor elevations and move Upper Shutter to telescope elevation.

D. Subsystem FILTERS: Filter slides and Camera Shutter control

4-1. FILTERS – LIMIT_FILTER				
<SUBSYSTEM>	"FILTERS"			
<COMMAND>	"LIMIT_FILTER"			
<RESPONSE>	<FL1> <FL2> <FL3> <FL4>			
Parameter	- <FL1> ~ <FL4>: Status of both Inner limit and Outer limit switch for all filter slides	Outer limit	Inner limit	Status <FL>
		OFF	OFF	"0"
		ON	OFF	"1"
		OFF	ON	"2"
		ON	ON	"3"
Function	Monitoring the limit status of all filter slides			
Example	KMTNET AUX 01 FILTERS LIMIT_FILTER KMTNET AUX 01 1 1 1 1 KMTNET AUX 02 FILTERS SET_F1 IN KMTNET AUX 02 OK KMTNET AUX 03 FILTERS LIMIT_FILTER KMTNET AUX 03 0 1 1 1 KMTNET AUX 04 FILTERS LIMIT_FILTER			

	KMTNET AUX 04 2 1 1 1 KMTNET AUX 05 FILTERS SET_F1 OUT KMTNET AUX 05 OK KMTNET AUX 06 FILTERS LIMIT_FILTER KMTNET AUX 06 0 1 1 1 KMTNET AUX 07 FILTERS SET_F2 IN KMTNET AUX 07 OK KMTNET AUX 08 FILTERS LIMIT_FILTER KMTNET AUX 08 0 0 1 1 KMTNET AUX 09 FILTERS LIMIT_FILTER KMTNET AUX 09 1 0 1 1 KMTNET AUX 10 FILTERS LIMIT_FILTER KMTNET AUX 10 1 2 1 1
Notes	- Inner limit switch is changed to ON when a filter slide is completely inserted between lenses and it is on optical path. - Outer limit switch is changed to ON when a filter slide is completely removed from optical path.

4-2. FILTERS – LIMIT_SHUT				
<SUBSYSTEM>	"FILTERS"			
<COMMAND>	"LIMIT_SHUT"			
<RESPONSE>	<SL1> <SL2>			
Parameter	- <SL1>: Limit status for the lower shutter blade (Full, M1 side) - <SL2>: Limit status for the upper shutter blade (Half, Camera side)	Outer limit	Inner limit	Status <SL>
		OFF	OFF	"0"
		ON	OFF	"1"
		OFF	ON	"2"
		ON	ON	"3"
Function	Monitoring the limit status of both shutter blades			
Example	If exposure time > shutter opening time, KMTNET AUX 01 FILTERS LIMIT_SHUT KMTNET AUX 01 2 1 (The camera shutters are on standby.) KMTNET AUX 02 FILTERS LIMIT_SHUT KMTNET AUX 02 0 1 (camera Full shutter opening - started) KMTNET AUX 03 FILTERS LIMIT_SHUT KMTNET AUX 03 1 1 (camera Full shutter opening - finished) KMTNET AUX 04 FILTERS LIMIT_SHUT KMTNET AUX 04 1 0 (camera Half shutter closing - started)			

	KMTNET AUX 05 FILTERS LIMIT_SHUT KMTNET AUX 05 1 2 (camera Half shutter closing - finished) KMTNET AUX 06 FILTER LIMIT_SHUT KMTNET AUX 06 0 0 (both Full and Half shutter return to standby position - started) KMTNET AUX 07 FILTER LIMIT_SHUT KMTNET AUX 07 2 1 (both Full and Half shutter return to standby position - finished) (The camera shutters are on standby.) If exposure time < shutter opening time, KMTNET AUX 11 FILTERS LIMIT_SHUT KMTNET AUX 11 2 1 (The camera shutters are on standby.) KMTNET AUX 12 FILTERS LIMIT_SHUT KMTNET AUX 12 0 1 (Only camera Full shutter opening - started) KMTNET AUX 13 FILTERS LIMIT_SHUT KMTNET AUX 13 0 0 (camera Full shutter opening & Half shutter closing - started) KMTNET AUX 14 FILTERS LIMIT_SHUT KMTNET AUX 14 1 0 (camera Full shutter opening - finished & Half shutter closing) KMTNET AUX 15 FILTERS LIMIT_SHUT KMTNET AUX 15 1 2 (camera Half shutter closing - finished) KMTNET AUX 16 FILTER LIMIT_SHUT KMTNET AUX 16 0 0 (both Full and Half shutter return to standby position - started) KMTNET AUX 07 FILTER LIMIT_SHUT KMTNET AUX 07 2 1 (both Full and Half shutter return to standby position - finished) (The camera shutters are on standby.)
Notes	- Inner limit switch is changed to ON when a shutter blade is completely inserted between lenses and it is on optical path. - Outer limit switch is changed to ON when a shutter blade is completely removed from optical path. If both Outer limits of Full Shutter and Half Shutter are ON, the shutter is completely open. - The camera shutter is controlled by the shutter input bit that is set through a hardware TTL signal input from the camera HE Box. - When the shutter input bit is set HIGH, starting camera exposure is started and camera Full shutter opening is started.

- When the shutter input bit is set LOW, finishing camera exposure is started and camera Half shutter closing is started.
- The following information can be referred for more details about the camera shutter operation and status monitor.

<< camera shutter operation and status info >>

- Shutter control input: Pin11 - TTL 5V input, pulled up / Pin10 - common
- OPEN CMD : LOW-->HIGH: The shutter starts opening when the input goes HIGH
- CLOSE CMD: HIGH-->LOW: The shutter starts closing when the input goes LOW
- NOTE: in the HE box utility operation for shutter open/close,
OPEN CMD should be commanded when the operational status is STANDBY
CLOSE CMD should be commanded when the operational status is OPENING or OPENED
- Limit status SF and SH
 - SF: limit status of Full shutter / SH: limit status of Half shutter
 - Limit status value: 0:no, 1:out, 2:in, 3:both (in-->block, out-->open)

- Table 1: Status and OpStatus configuration with Limit status SF/SH

CASE A: exposure time > shutter opening time

Input	--	Limit	--	Status	--	OpStatus	(duration time)	Remark
LOW	--	2 1	--	closed	--	standby		
HIGH	--	0 1	--	open	--	opening	(5 sec)	
HIGH	--	1 1	--	open	--	opened	(ExpTime - 5 sec)	
LOW	--	1 0	--	open	--	closing	(5 sec)	
LOW	--	1 2	--	closed	--	reloading	(0.5 sec)	
LOW	--	0 2	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	1 0	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	0 0	--	closed	--	reloading	(5 sec)	
LOW	--	2 0	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	0 1	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	2 1	--	closed	--	standby		

CASE B: exposure time < shutter opening time

Input	--	Limit	--	Status	--	OpStatus	(duration time)	Remark
LOW	--	2 1	--	closed	--	standby		
HIGH	--	0 1	--	open	--	opening	(ExpTime)	
LOW	--	0 0	--	open	--	opening	(5 sec - ExpTime)	& closing
LOW	--	1 0	--	open	--	closing	(ExpTime)	
LOW	--	1 2	--	closed	--	reloading	(0.5 sec)	
LOW	--	0 2	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	1 0	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	0 0	--	closed	--	reloading	(5 sec)	
LOW	--	2 0	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	0 1	--	closed	--	reloading	(0.1 sec)	possibly
LOW	--	2 1	--	closed	--	standby		

- Table 2: Lookup table for Status/OpStatus with SF/SH

SF	SH	Status / OpStatus	previous OpStatus or SF SH
2	1	-->	closed / standby

	1 1 --> open / opened 0 1 --> open / opening standby* or (opening) --> closed / reloading reloading* or (reloading) 1 0 --> open / closing opened* or opening* or (closing) --> closed / reloading reloading* or closing or (reloading) 0 0 --> closed / reloading 1 2* or 1 0* or 1 0 or 0 2 or (0 0) --> open / opening 0 1* or 2 1* or (0 0) 1 2 --> closed / reloading 0 2 --> closed / reloading 2 0 --> closed / reloading -----
--	--

4-3. FILTERS – SET_F1 ~ SET_F4	
<SUBSYSTEM>	"FILTERS"
<COMMAND>	"SET_F1" <Action> "SET_F2" <Action> "SET_F3" <Action> "SET_F4" <Action>
<RESPONSE>	<ACK> / <WAIT>
Parameter	- <Action> - "IN" : The filter slide will be inserted between lenses. - "OUT" : The filter slide will be removed from optical path.
Function	Separate In/Out control for each filter slide
Example	KMTNET AUX 01 FILTERS LIMIT_FILTER KMTNET AUX 01 2 1 1 1 KMTNET AUX 02 FILTERS SET_F1 OUT KMTNET AUX 02 OK KMTNET AUX 03 FILTERS STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 FILTERS LIMIT_FILTER KMTNET AUX 04 0 1 1 1 KMTNET AUX 05 FILTERS SET_F2 IN KMTNET AUX 05 OK KMTNET AUX 06 FILTERS STATUS KMTNET AUX 06 RUNNING KMTNET AUX 07 FILTERS LIMIT_FILTER KMTNET AUX 07 0 0 1 1 KMTNET AUX 08 FILTERS STATUS KMTNET AUX 08 RUNNING KMTNET AUX 09 FILTERS STATUS KMTNET AUX 09 STANDBY KMTNET AUX 10 FILTERS LIMIT_FILTER KMTNET AUX 10 1 2 1 1 KMTNET AUX 11 FILTERS SET_F2 OUT KMTNET AUX 11 OK KMTNET AUX 12 FILTERS SET_F2 IN KMTNET AUX 12 WAIT KMTNET AUX 13 FILTERS STATUS KMTNET AUX 13 RUNNING

	KMTNET AUX 14 FILTERS STATUS KMTNET AUX 14 RUNNING KMTNET AUX 15 FILTERS STATUS KMTNET AUX 15 STANDBY KMTNET AUX 16 FILTERS LIMIT_FILTER KMTNET AUX 16 1 1 1 1 KMTNET AUX 17 FILTERS SET_F2 IN KMTNET AUX 17 OK KMTNET AUX 18 FILTERS STATUS KMTNET AUX 18 RUNNING KMTNET AUX 19 FILTERS STATUS KMTNET AUX 19 STANDBY KMTNET AUX 16 FILTERS LIMIT_FILTER KMTNET AUX 16 1 2 1 1
Notes	- Two or more filter slides can be moved at the same time. - If the commanded filter slide is still running since previous operation is not completed yet, <WAIT> is returned without starting the operation. - SET_Fx operation is able to start regardless of camera shutter operation.

4-4. FILTERS –FNAMES	
<SUBSYSTEM>	"FILTERS"
<COMMAND>	"FNAMES "
<RESPONSE>	<FN1> <FN2> <FN3> <FN4>
Parameter	- <FN1> ~ <FN4>: Filter names for each slide
Function	Return the filter set information
Example	KMTNET AUX 01 FILTERS FNAMES KMTNET AUX 01 I R V B
Notes	-

E. Subsystem M1COVER: Primary Mirror Cover control

5-1. M1COVER – POSITION	
<SUBSYSTEM>	"M1COVER"
<COMMAND>	"POSITION"
<RESPONSE>	<Mirror cover position>
Parameter	- <Mirror cover position>: Current position in percentage, integer number 0~100. → ex) "100"

Function	Monitoring the Mirror Cover position
Example	KMTNET AUX 01 M1COVER POSITION KMTNET AUX 01 0 KMTNET AUX 02 M1COVER OPEN KMTNET AUX 02 OK KMTNET AUX 03 M1COVER STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 M1COVER POSITION KMTNET AUX 04 34 KMTNET AUX 05 M1COVER POSITION KMTNET AUX 05 58 KMTNET AUX 06 M1COVER POSITION KMTNET AUX 06 82 KMTNET AUX 07 M1COVER POSITION KMTNET AUX 07 100 KMTNET AUX 08 M1COVER STATUS KMTNET AUX 08 STANDBY
Notes	-

5-2. M1COVER – OPEN/CLOSE	
<SUBSYSTEM>	"M1COVER"
<COMMAND>	"OPEN" "CLOSE"
<RESPONSE>	<ACK> / <WAIT>
Parameter	-
Function	Mirror Cover Open/Close
Example	KMTNET AUX OP1 M1COVER POSITION KMTNET AUX OP1 0 KMTNET AUX OP2 M1COVER OPEN KMTNET AUX OP2 OK KMTNET AUX OP3 M1COVER STATUS KMTNET AUX OP3 RUNNING KMTNET AUX OP4 M1COVER STATUS KMTNET AUX OP4 STANDBY KMTNET AUX OP5 M1COVER POSITION KMTNET AUX OP5 100 KMTNET AUX CL1 M1COVER POSITION KMTNET AUX CL1 100 KMTNET AUX CL2 M1COVER CLOSE KMTNET AUX CL2 OK KMTNET AUX CL3 M1COVER STATUS KMTNET AUX CL3 RUNNING KMTNET AUX CL4 M1COVER STATUS KMTNET AUX CL4 STANDBY KMTNET AUX CL5 M1COVER POSITION KMTNET AUX CL5 0
Notes	- If Mirror Cover is running for opening/closing,

	OPEN/CLOSE operation is not started and <WAIT> is returned.
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5-3. M1COVER – STOP	
<SUBSYSTEM>	"M1COVER"
<COMMAND>	"STOP"
<RESPONSE>	<ACK>
Parameter	-
Function	Mirror Cover operation stop
Example	<pre> KMTNET AUX 01 M1COVER POSITION KMTNET AUX 01 0 KMTNET AUX 02 M1COVER OPEN KMTNET AUX 02 OK KMTNET AUX 03 M1COVER STATUS KMTNET AUX 03 RUNNING KMTNET AUX 04 M1COVER CLOSE KMTNET AUX 04 WAIT KMTNET AUX 05 M1COVER STATUS KMTNET AUX 05 RUNNING KMTNET AUX 06 M1COVER STOP KMTNET AUX 06 OK KMTNET AUX 07 M1COVER STATUS KMTNET AUX 07 STANDBY KMTNET AUX 08 M1COVER POSITION KMTNET AUX 08 42 KMTNET AUX 09 M1COVER CLOSE KMTNET AUX 09 OK KMTNET AUX 10 M1COVER STATUS KMTNET AUX 10 RUNNING KMTNET AUX 11 M1COVER STATUS KMTNET AUX 11 STANDBY KMTNET AUX 12 M1COVER POSITION KMTNET AUX 12 0 </pre>
Notes	-

F. Subsystem CHILLER and ENVIRON: Mirror cooling control and Environment monitoring

6-1. CHILLER – GET_TEMP	
<SUBSYSTEM>	"CHILLER"
<COMMAND>	"GET_TEMP"
<RESPONSE>	<Process temperature>

Parameter	- <Process temperature>: Current process temperature of the coolant, floating-point number including a sign in °C. ex) "+14.2"
Function	Monitoring the processed temperature of Chiller
Example	KMTNET AUX C001 CHILLER GET_TEMP KMTNET AUX C001 +19.2
Notes	-

6-2. CHILLER – SET_SETPOINT /GET_SETPOINT	
<SUBSYSTEM>	"CHILLER"
<COMMAND>	"SET_SETPOINT" <Set point> "GET_SETPOINT"
<RESPONSE>	(SET) <ACK> (GET) <Set point>
Parameter	- <Set point>: Set point temperature, floating-point number including a sign in °C. → ex) "+14.0"
Function	Setting and Monitoring the set point temperature of Chiller
Example	KMTNET AUX C001 CHILLER GET_SETPOINT KMTNET AUX C001 +10.0 KMTNET AUX C002 CHILLER SET_SETPOINT 5.0 KMTNET AUX C002 OK KMTNET AUX C003 CHILLER GET_SETPOINT KMTNET AUX C003 +5.0
Notes	-

6-3. CHILLER – SET_POWER/GET_POWER	
<SUBSYSTEM>	"CHILLER"
<COMMAND>	"SET_POWER" <Action> "GET_POWER"
<RESPONSE>	(SET) <ACK> (GET) <Chiller power>
Parameter	- <Action > → "ON" / "OFF" - <Chiller power> → "ON" / "OFF"
Function	Chiller power On/Off control and monitoring
Example	KMTNET AUX C001 CHILLER GET_POWER KMTNET AUX C001 OFF KMTNET AUX C002 CHILLER SET_POWER ON KMTNET AUX C002 OK

	KMTNET AUX C003 CHILLER GET_POWER KMTNET AUX C003 ON
Notes	-

6-4. ENVIRON – SENSORS	
<SUBSYSTEM>	"ENVIRON"
<COMMAND>	"SENSORS"
<RESPONSE>	<SE1> <SE2> <SE3> <SE4> <SE5> <SE6> <SE7>
Parameter	<SE1>: Ambient air temperature at the Head Ring, °C <SE2>: Corrector Box(south) temperature, °C <SE3>: OTA Center-East temperature, °C <SE4>: Humidity at the Head Ring, RH % <SE5>: Primary Mirror NW edge temperature, °C <SE6>: Air temperature inside Mirror Cell(NW side), °C <SE7>: Reserved, °C or RH %
Function	Monitoring the Environment with the temperature or humidity sensors
Example	KMTNET AUX E001 ENVIRON SENSORS KMTNET AUX E001 13.2 13.6 13.7 20.4 15.5 15.6 0.0
Notes	-

6-5. ENVIRON – SET_FAN/GET_FAN	
<SUBSYSTEM>	"ENVIRON"
<COMMAND>	"SET_FAN" <Action> "GET_FAN"
<RESPONSE>	(SET) <ACK> (GET) <Fan power>
Parameter	- <Action> → "ON" / "OFF" - <Fan power> → "ON" / "OFF"
Function	Mirror cooling fan On/Off control and monitoring
Example	KMTNET AUX E001 ENVIRON GET_FAN KMTNET AUX E001 OFF KMTNET AUX E002 ENVIRON SET_FAN ON KMTNET AUX E002 OK KMTNET AUX E003 ENVIRON GET_FAN KMTNET AUX E003 ON
Notes	-

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